

ROCK OF AGES, MI, USA

WMO: 994130

Lat: 47.867N

Lon: 89.313W

Elev: 184

StdP: 99.14

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.2	(c) -20.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 20.9	(k) -9.6	(l) 19.0	(m) -7.3	(n) 8.6	(o) 340	(p) 0.925

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.7	(c) 22.9	(d) N/A	(e) 21.5	(f) N/A	(g) 20.2	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 7.7	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 18.4	(b) 16.2	(c) 14.7	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) 18.4	(b) 16.2	(c) 14.7	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.5	-9.4	-8.2	-2.4	2.9	7.4	11.8	15.8	16.6	13.9	7.7	1.2	-4.2
	DBStd	9.80	6.82	5.86	5.44	4.02	3.09	2.55	3.16	2.63	3.23	3.59	4.86	5.23
	HDD10.0	2609	600	509	385	214	92	10	1	0	6	88	263	441
	HDD18.3	5077	858	742	643	463	338	196	91	67	137	329	513	699
	CDD10.0	600	0	0	0	0	12	64	179	204	122	18	0	0
	CDD18.3	26	0	0	0	0	0	0	11	13	2	0	0	0
	CDH23.3	32	0	0	0	0	0	5	18	8	1	0	0	0
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0
(14) Wind	WSAvg	7.6	9.2	8.3	7.7	7.3	7.1	6.2	6.2	6.5	7.2	8.4	8.7	9.0
(15) Precipitation	PrecAvg	731	47	31	39	63	74	85	77	68	82	80	62	47
	PrecMax	902	94	59	70	188	112	150	139	114	168	200	114	80
	PrecMin	591	18	16	16	20	41	22	26	28	31	31	24	14
	PrecStd	80	17	10	16	36	25	33	32	26	33	41	24	17
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	5.0	10.4	13.9	19.8	23.7	25.7	24.4	22.5	18.0	10.8	5.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	2.8	2.2	7.3	11.4	16.0	20.1	23.0	22.7	20.3	15.6	9.1	4.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	1.6	0.7	5.0	9.6	14.1	18.1	21.5	21.3	19.0	13.8	8.1	3.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	0.1	-0.6	3.4	8.0	12.3	16.4	20.2	20.2	17.8	12.1	7.2	2.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	6.0	5.4	5.5	6.6	8.2	7.1	4.7	4.3	3.9	4.3	5.3
		MCDBR	5.9	5.3	6.5	8.4	10.1	11.2	8.4	6.2	6.0	6.0	4.2	4.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.267	0.279	0.305	0.343	0.366	0.384	0.396	0.378	0.360	0.322	0.294	0.271
	taud		2.471	2.417	2.398	2.365	2.398	2.368	2.400	2.457	2.480	2.541	2.532	2.493
	Ebn at Noon		832	895	919	909	897	878	862	865	849	834	791	782
	Edh at Noon		62	82	98	113	114	118	113	102	91	73	60	55
(44) All-Sky Solar Radiation	RadAvg		1.00	1.98	3.40	4.70	5.52	6.00	6.07	5.21	3.75	2.21	1.12	0.77
	RadStd		0.11	0.23	0.32	0.51	0.37	0.40	0.29	0.37	0.23	0.27	0.14	0.09

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

[Nomenclature: See separate page](#)